

Store Network Reference

Internal IT Documentation — Cisco Meraki Standard Configuration

1. Overview

Every retail store is deployed with a standardized Cisco Meraki network stack: one MX firewall (gateway), one MS switch (48-port), and one MR access point. All stores share the same VLAN architecture and IP addressing scheme.

Device	Model	Role
Firewall	MX67	Gateway, DHCP, FW, IDS
Switch	MS130-48P	Layer 2, PoE, QoS
Access Point	MR44	Dual-band WiFi

2. IP Addressing Scheme

Format: 10.{store_number}.{vlan_id}.0/24

- VLAN 10 → 10.{store}.10.0/24, gateway 10.{store}.10.1
- VLAN 20 → 10.{store}.20.0/24, gateway 10.{store}.20.1

Stores with numbers above 255 map to IP octet 200.

3. VLAN Architecture

VLAN	Name	Purpose	DHCP Lease
10	Store	Management PC, Printer	12 hours
20	Toast_POS	Toast terminals, POS printers	12 hours
30	Voice	VoIP phones	12 hours
40	Verkada	Security cameras	12 hours
50	Digital_Signage	Menu boards	12 hours
60	Kitchen_IoT	Kitchen equipment	12 hours
91	WiFi_Staff	Staff WiFi SSID	1 hour
100	WiFi_Guest	Guest WiFi SSID	30 minutes

4. Wireless SSIDs

SSID	VLAN	Auth	Visible	Notes
Guest	100	Open	Yes	Splash page, bandwidth limited
Staff	91	WPA2-PSK	Hidden	Staff only

5. Switch Port Layout (48-port standard)

SW Port	Name	VLAN	Type	STP Guard
1-2	TOAST-POS-1/2	20	access	root guard
3-4	TOAST-PRINTER-1/2	20	access	root guard
5-6	POS-SPARE	20	access	root guard
7	VOIP-PHONE	30	access	root guard
8	STORE-PRINTER	10	access	root guard
9	OTTER-3P-PRINTER	60	access	root guard
11-12	DLVR-PC / SPARE-PC	10	access	root guard
13	RATIONAL-OVEN	60	access	root guard
15-24	VERKADA-CAM-1 to 10	40	access	root guard
25-26	AP-UPLINK	10	trunk	disabled
27-33	SIGNAGE-1 to 7	50	access	root guard
47	MUSIC-PLAYER	50	access	root guard
48	MX-UPLINK	10	trunk	disabled

6. Firewall Rules (Inter-VLAN Isolation)

Source	Destination	Action	Notes
Staff WiFi (91)	Store VLAN (10)	ALLOW	Staff WiFi to Store VLAN
Staff WiFi (91)	All internal	DENY	Blocked from all else
Guest (100)	All internal	DENY	Guest isolated
Toast POS (20)	All internal	DENY	POS isolated
Voice (30)	All internal	DENY	Voice isolated
Verkada (40)	All internal	DENY	Camera isolated
Signage (50)	All internal	DENY	Signage isolated
Kitchen IoT (60)	All internal	DENY	IoT isolated

Sanitized for portfolio use — internal identifiers removed